

ADVANCED

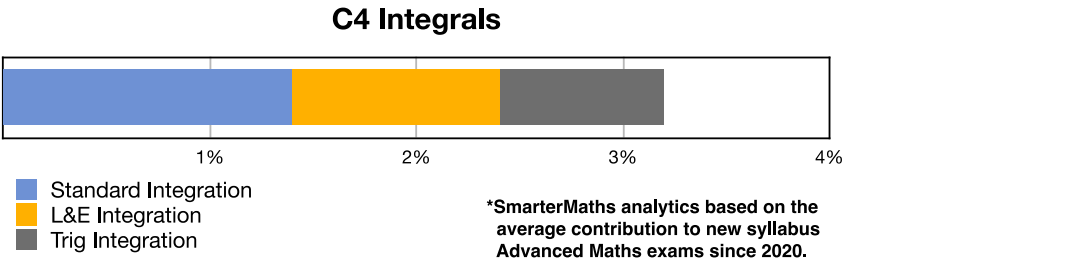
Stage 6

Calculus (Adv), C4 Integration (Adv), Integration (Y12)

Trig Integration

Teacher: SHS Maths

Exam Equivalent Time: 55.5 minutes (based on allocation of 1.5 minutes per mark)



HISTORICAL CONTRIBUTION

- C4 Integration* has contributed a substantial 14.0% to new syllabus Advanced exams since it was introduced in 2020.
- This topic has been split into four sub-topics for analysis purposes: *1-Integrals (3.2%)*, *2-Areas Under Curves (5.8%)*, *3-Trapezoidal Rule (2.2%)* and *4-Other Integration Examples (2.8%)*.
- This analysis looks at *Integrals*.

HSC ANALYSIS - What to expect and common pitfalls

- Integrals (3.2%)* is made up of standard integration, log and exponential integration and trig integration.
- Trig Integration* was most recently examined in 2024 in a difficult *differentiate then integrate* question that is a must review question!
- Before 2024, it was tested in 2020 in a band 3 questions that was very well answered. We note that this area is typically tested at the slightly higher band 4 difficulty level and many examples in the database cover this.

Questions

1. Calculus, 2ADV C4 2019 HSC 9 MC

Which expression is equal to $\int \tan^2 x \, dx$?

- A.

$\tan x - x + C$
- B.

$\tan x - 1 + C$
- C.

$\frac{\tan^3 x^2}{6} + C$
- D.

$\frac{\tan^3 x}{3} + C$

2. Calculus, 2ADV C4 2020 HSC 13

Evaluate $\int_0^{\frac{\pi}{4}} \sec^2 x \, dx$. *(2 marks)*

3. Calculus, 2ADV C4 2008 HSC 3b

- i.

Differentiate $\log_e(\cos x)$ with respect to x . *(2 marks)*
- ii.

Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{4}} \tan x \, dx$. *(2 marks)*

4. Calculus, 2ADV C4 2007 HSC 2bi

Find $\int (1 + \cos 3x) \, dx$. *(2 marks)*

5. Calculus, 2ADV C4 2023 MET1 5

- a.

Evaluate $\int_0^{\frac{\pi}{3}} \sin(x) \, dx$. *(1 mark)*
- b.

Hence, or otherwise, find all values of k such that $\int_0^{\frac{\pi}{3}} \sin(x) \, dx = \int_k^{\frac{\pi}{2}} \cos(x) \, dx$, where $-3\pi < k < 2\pi$. *(3 marks)*

6. Calculus, 2ADV C4 2008 HSC 5a

The gradient of a curve is given by $\frac{dy}{dx} = 1 - 6\sin 3x$. The curve passes through the point $(0, 7)$.

What is the equation of the curve? *(3 marks)*

7. Calculus, 2ADV C4 2012 HSC 11g

Find $\int_0^{\frac{\pi}{2}} \sec^2\left(\frac{x}{2}\right) dx$ (3 marks)

8. Calculus, 2ADV C4 2014 HSC 11e

Evaluate $\int_0^{\frac{\pi}{2}} \sin\left(\frac{x}{2}\right) dx$. (3 marks)

9. Calculus, 2ADV C4 2014 HSC 13a

- i. Differentiate $3 + \sin 2x$. (1 mark)
- ii. Hence, or otherwise, find $\int \frac{\cos 2x}{3 + \sin 2x} dx$. (2 marks)

10. Calculus, 2ADV C4 2008 HSC 2cii

Evaluate $\int_0^{\frac{\pi}{12}} \sec^2 3x dx$. (3 marks)

11. Calculus, 2ADV C4 2024 HSC 27

- a. Find the derivative of $x^2 \tan x$ (2 marks)
- b. Hence, find $\int (x \tan x + 1)^2 dx$ (3 marks)

12. Calculus, 2ADV C4 2010 HSC 5b

- i. Prove that $\sec^2 x + \sec x \tan x = \frac{1 + \sin x}{\cos^2 x}$. (1 mark)
- ii. Hence prove that $\sec^2 x + \sec x \tan x = \frac{1}{1 - \sin x}$. (1 mark)
- iii. Hence, use the identity $\int \sec x \tan x dx = \frac{1}{a} \sec ax$ to find the exact value of $\int_0^{\frac{\pi}{4}} \frac{1}{1 - \sin x} dx$. (2 marks)

Worked Solutions

1. Calculus, 2ADV C4 2019 HSC 9 MC

Consider option A:

$$\frac{d}{dx}(\tan x - x + C)$$

$$= \sec^2 x - 1$$

$$= \tan^2 x$$

$$\therefore \int \tan^2 x dx = \tan x - x + C$$

$$\Rightarrow A$$

♦♦ Mean mark 35%.

2. Calculus, 2ADV C4 2020 HSC 13

$$\begin{aligned} \int_0^{\frac{\pi}{4}} \sec^2 x &= [\tan x]_0^{\frac{\pi}{4}} \\ &= \tan \frac{\pi}{4} - \tan 0 \\ &= 1 \end{aligned}$$

3. Calculus, 2ADV C4 2008 HSC 3b

i. $y = \log_e(\cos x)$

$$\begin{aligned} \frac{dy}{dx} &= \frac{-\sin x}{\cos x} \\ &= -\tan x \end{aligned}$$

ii. $\int_0^{\frac{\pi}{4}} \tan x dx$

$$= -[\log_e(\cos x)]_0^{\frac{\pi}{4}}$$

$$= -\left[\log_e\left(\cos\left(\frac{\pi}{4}\right)\right) - \log_e(\cos 0)\right]$$

$$= -\left[\log_e\left(\frac{1}{\sqrt{2}}\right) - \log_e 1\right]$$

$$= -\left[\log_e\left(\frac{1}{\sqrt{2}}\right) - 0\right]$$

$$= -\log_e\left(\frac{1}{\sqrt{2}}\right)$$

$$= 0.346\dots$$

$$= 0.35 \text{ (2 d.p.)}$$

4. Calculus, 2ADV C4 2007 HSC 2bi

$$\begin{aligned}\int (1 + \cos 3x) \, dx \\ = x + \frac{1}{3} \sin 3x + C\end{aligned}$$

5. Calculus, 2ADV C4 2023 MET1 5

$$\begin{aligned}\text{a. } \int_0^{\frac{\pi}{3}} \sin(x) \, dx &= [-\cos x]_0^{\frac{\pi}{3}} \\ &= -\cos \frac{\pi}{3} + \cos 0 \\ &= -\frac{1}{2} + 1 \\ &= \frac{1}{2}\end{aligned}$$

$$\begin{aligned}\text{b. } \int_k^{\frac{\pi}{2}} \cos(x) \, dx &= [\sin x]_k^{\frac{\pi}{2}} \\ &= \sin\left(\frac{\pi}{2}\right) - \sin(k) \\ &= 1 - \sin(k)\end{aligned}$$

Using part (a):

$$\begin{aligned}1 - \sin(k) &= \frac{1}{2} \\ \sin(k) &= \frac{1}{2} \\ \therefore k &= \frac{-11\pi}{6}, \frac{-7\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}\end{aligned}$$

6. Calculus, 2ADV C4 2008 HSC 5a

$$\begin{aligned}\frac{dy}{dx} &= 1 - 6\sin 3x \\ y &= \int 1 - 6\sin 3x \, dx \\ &= x + 2\cos 3x + c\end{aligned}$$

Passes through $(0, 7)$:

$$\begin{aligned}\Rightarrow 0 + 2\cos 0 + c &= 7 \\ 2 + c &= 7 \\ c &= 5\end{aligned}$$

$\therefore$ Equation is $y = x + 2\cos 3x + 5$

7. Calculus, 2ADV C4 2012 HSC 11g

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sec^2\left(\frac{x}{2}\right) \, dx \\ = \left[2\tan\left(\frac{x}{2}\right)\right]_0^{\frac{\pi}{2}} \\ = 2\tan \frac{\pi}{4} - 2\tan 0 \\ = 2(1) - 0 \\ = 2\end{aligned}$$

8. Calculus, 2ADV C4 2014 HSC 11e

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin\left(\frac{x}{2}\right) \, dx \\ = \left[-2\cos\left(\frac{x}{2}\right)\right]_0^{\frac{\pi}{2}} \\ = -2\left[\cos\left(\frac{\pi}{4}\right) - \cos 0\right] \\ = -2\left[\frac{1}{\sqrt{2}} - 1\right] \\ = -\frac{2}{\sqrt{2}} + 2 \\ = 2 - \sqrt{2}\end{aligned}$$

9. Calculus, 2ADV C4 2014 HSC 13a

i. $y = 3 + \sin 2x$

$$\frac{dy}{dx} = 2\cos 2x$$

ii.

$$\int \frac{\cos 2x}{3 + \sin 2x} \, dx$$
$$= \frac{1}{2} \int \frac{2\cos 2x}{3 + \sin 2x} \, dx$$
$$= \frac{1}{2} \ln(3 + \sin 2x) + C \quad (\text{from part (i)})$$

10. Calculus, 2ADV C4 2008 HSC 2cii

$$\int_0^{\frac{\pi}{12}} \sec^2 3x \, dx$$
$$= \left[\frac{1}{3} \tan 3x \right]_0^{\frac{\pi}{12}}$$
$$= \left[\left(\frac{1}{3} \tan \left(\frac{\pi}{4} \right) \right) - \left(\frac{1}{3} \tan 0 \right) \right]$$
$$= \frac{1}{3} (1) - 0$$
$$= \frac{1}{3}$$

11. Calculus, 2ADV C4 2024 HSC 27

a. $y = x^2 \tan x$

By product rule:

$$\frac{dy}{dx} = 2x \tan x + x^2 \sec^2 x$$

b. $\int (x \tan x + 1)^2 \, dx$

$$= \int x^2 \tan^2 x + 2x \tan x + 1 \, dx$$
$$= \int x^2 (\sec^2 x - 1) + 2x \tan x + 1 \, dx$$
$$= \int 2x \tan x + x^2 \sec^2 x - x^2 + 1 \, dx$$
$$= x^2 \tan x - \frac{x^3}{3} + x + C$$

♦♦ Mean mark (b) 34%.

12. Calculus, 2ADV C4 2010 HSC 5b

i. Need to prove

$$\sec^2 x + \sec x \tan x = \frac{1 + \sin x}{\cos^2 x}$$

LHS = $\sec^2 x + \sec x \tan x$

$$= \frac{1}{\cos^2 x} + \frac{1}{\cos x} \times \frac{\sin x}{\cos x}$$
$$= \frac{1}{\cos^2 x} + \frac{\sin x}{\cos^2 x}$$
$$= \frac{1 + \sin x}{\cos^2 x}$$

= RHS ... as required

ii. Need to prove

$$\sec^2 x + \sec x \tan x = \frac{1}{1 - \sin x}$$

i.e. $\frac{1 + \sin x}{\cos^2 x} = \frac{1}{1 - \sin x} \quad (\text{part (i)})$

LHS = $\frac{1 + \sin x}{\cos^2 x}$

$$= \frac{1 + \sin x}{1 - \sin^2 x}$$
$$= \frac{1 + \sin x}{(1 - \sin x)(1 + \sin x)}$$
$$= \frac{1}{1 - \sin x} \quad \dots \text{as required}$$

♦♦ Mean mark 31%.

iii. $\int_0^{\frac{\pi}{4}} \frac{1}{1 - \sin x} \, dx$

$$= \int_0^{\frac{\pi}{4}} (\sec^2 x + \sec x \tan x) \, dx$$
$$= [\tan x + \sec x]_0^{\frac{\pi}{4}}$$
$$= \left[\left(\tan \left(\frac{\pi}{4} \right) + \sec \left(\frac{\pi}{4} \right) \right) - (\tan 0 + \sec 0) \right]$$
$$= \left[\left(1 + \frac{1}{\cos \left(\frac{\pi}{4} \right)} \right) - \left(0 + \frac{1}{\cos 0} \right) \right]$$
$$= 1 + \sqrt{2} - 1$$
$$= \sqrt{2}$$

♦ Mean mark 37%.

